

$$\sum_{n=0}^{\infty} [b'_n(t) + b_n(t)(n\pi)^2] \sin(n\pi x) = \sum_{n=0}^{\infty} e^{-t} c_n \sin(n\pi x)$$

$$\Rightarrow b'_n(t) + b_n(t)(n\pi)^2 = e^{-t} c_n$$

* We applied our B.C. to find $X_n(x)$, we have it applied I.C.: Do that here:

I.C.: • $u(x, 0) = 1$

• $u(x, t) = \sum_{n=0}^{\infty} b_n(t) \sin(n\pi x)$

$$\Rightarrow u(x, 0) = \sum_{n=0}^{\infty} b_n(0) \sin(n\pi x) = 1$$

Recall: $1 = \sum_{n=0}^{\infty} c_n \sin(n\pi x)$

$\Rightarrow$

* $b_n(0) = c_n$

$$\Rightarrow u(x, t) = \sum_{n=0}^{\infty} b_n(t) \sin(n\pi x)$$

(i) apply I.C. $u(x, 0) = \sum_{n=0}^{\infty} b_n(0) \sin(n\pi x) = 1$

$\underbrace{\quad}_{t\text{-only}} \quad \underbrace{\quad}_{x\text{-only}}$

$$\sum_{n=0}^{\infty} b_n(0) \sin(n\pi x) = \sum_{n=0}^{\infty} c_n \sin(n\pi x)$$

* $b_n(0) = c_n$ * ODE, I.C.

* $b'_n(t) + b_n(t)(n\pi)^2 = c_n e^{-t}$ *

Solve ODE: int. factor

• $\frac{d}{dt} (b_n(t) g(t)) = c_n e^{-t} g(t)$

$$b'_n g(t) + b_n(t) g'(t) = c_n e^{-t} g(t)$$

$$\Rightarrow \text{ODE: } g'(t) = (n\pi)^2 g(t)$$

$$g(t) = \exp((n\pi)^2 t)$$

$$* b_n(t) e^{(n\pi)^2 t} = \frac{c_n \exp[(n\pi)^2 - 1)t]}{(n\pi)^2 - 1} + \tilde{c}_n$$

$$c_n e^{-t} + \tilde{c}_n e^{(n\pi)^2 t}$$

$$b_n(t) = \frac{c_n e^{-t}}{(n\pi)^2 - 1}$$

now apply our I.C.

$$* b_n(0) = c_n$$

$$b_n(0) = \frac{c_n}{(n\pi)^2 - 1} + \hat{c}_n = c_n$$

$$\leadsto \text{solve for } \hat{c}_n: \quad \hat{c}_n = c_n \left[1 - \frac{1}{(n\pi)^2 - 1} \right]$$

$$= c_n \left[\frac{(n\pi)^2 - 2}{(n\pi)^2 - 1} \right]$$

$$b_n(t) = \frac{c_n e^{-t}}{(n\pi)^2 - 1} + c_n \left[\frac{(n\pi)^2 - 2}{(n\pi)^2 - 1} \right] e^{-(n\pi)^2 t}$$

$$= \frac{c_n}{((n\pi)^2 - 1)} \left(e^{-t} + ((n\pi)^2 - 2) e^{-(n\pi)^2 t} \right)$$

$$* \text{ Plug back into } u(x, t) = \sum_{n=0}^{\infty} b_n(t) \sin(n\pi x)$$

$$u(x, t) = \sum_{n=0}^{\infty} \frac{c_n}{((n\pi)^2 - 1)} \left(e^{-t} + ((n\pi)^2 - 2) e^{-(n\pi)^2 t} \right) \sin(n\pi x)$$

(PDE solⁿ)

$$u_t - u_{xx} = e^{-t}$$

$$* \text{ Find } c_n: \quad 1 = \sum_{n=0}^{\infty} c_n \sin(n\pi x)$$

by orthogonal polynomials

(i) multiply by $\sin(m\pi x)$

$$\sin(m\pi x) = \sum_{n=0}^{\infty} c_n \sin(m\pi x) \sin(n\pi x)$$

(ii) Integrate over $0 < x < 1$

$$\int_0^1 \sin(m\pi x) dx = \sum_{n=0}^{\infty} c_n \int_0^1 \sin(m\pi x) \sin(n\pi x) dx$$

$$\int_0^1 \sin(m\pi x) dx = \frac{c_n}{2}$$

$$c_n = 2 \int_0^1 \sin(m\pi x) dx$$

$$c_n = \frac{2 [1 - (-1)^n]}{(n\pi)}$$

Quiz 17: $u_{tt} = u_{xx} + \cos(x) \sin(t)$, $0 < x < \pi$

$u_x(0, t) = 0 = u_x(\pi, t)$ $t \geq 0$

$u(x, 0) = u_t(x, 0) = 0$

PDE: $u_{tt} - u_{xx} = \cos(x) \sin(t)$ (Inhom) $\neq 0$

* [Hom.] $u_{tt} - u_{xx} = 0$

$u(x, t) = X(x) T(t)$

Two ODE: $X''(x) = -\lambda^2 X$ $\leftarrow$ Eigen f_s^u
 ~~$T'(t) = -\lambda^2 T$~~

$X(x) = A \cos(\lambda x) + B \sin(\lambda x)$

B.C.: $u_x(0, t) = 0 \Rightarrow X'(0) = 0$

$u(\pi, t) = 0 \Rightarrow X(\pi) = 0$

$X'(x) = -A \lambda \sin(\lambda x) + B \lambda \cos(\lambda x)$

$X'(0) = B \lambda = 0 \Rightarrow B = 0$

$X(x) = A \cos(\lambda x)$

$X(\pi) = A \cos(\lambda \pi) = 0$

$\lambda_n \pi = \pi \left(\frac{1}{2} + n \right)$

$\lambda_n = \frac{1}{2} + n$

$X_n(x) = \cos\left(\left(\frac{1}{2} + n\right)x\right)$

$u(x, t) = \sum_{n=0}^{\infty} b_n(t) \cos\left[\left(\frac{1}{2} + n\right)x\right]$

